Faculty: Computers and Informatics **Class Year:** 2020/2021 **Year:** First Year **Subject Name:** Operations Research	**Mid-term Exam**	**Date:** June. 2021 **Time Allowed:** 30 Min. **No. of Pages:** 2 **No. of Questions:** 2 **Model :** 2

Answer the following questions: **(15 degree)**

Question 1:Mark (A) for true and (B) for false:

1. Variable costs per unit vary directly with levels of production. ✓
2. Total fixed costs vary inversely with levels of production. ✗ Fixed costs are constants
3. In operations research, the mathematical models are prepared for situations. ✓
4. If the right-hand side of a constraint is changed, the feasible region will not be affected and will remain the same. ✗ changes
5. A linear programming model consists of decision variables, constraints, but no objective function. ✗ it has an objective function
6. Slack variables are only associated with maximization problems. ✗ it may be in minimization problems
7. In the simplex method, a tableau is optimal only if all the cj– zj values are zero. ✗ <= 0
8. The constraint 5 X1 + 6 X2 = 30, when converted to an = constraint for use in the simplex algorithm, will be 5 X1 + 6 X2 + M = 30. ✗ the question is not written correctly but anyway the answer is false
9. Graphical optimal value for Z can be obtained from corner points of the solution region. ✓
10. Multiple optimal solutions cannot be determined from the simplex method. ✗

Question 2: Choose the correct answer:

11. The standard form of LPP if the objective function is of minimization then the right hand side of the constraints should be_____.
    a. positive .
    b. negative .
    c. non-negative .
    d. zero .

12. ABC company sells shoes for $450. The variable cost is $200 per unit. The fixed costs are $750,000. What is the breakeven in sales dollars?
    a. $750,000
    b. $937,500
    c. $1,350,000
    d. $1,687,500

$$V = \frac{C_F}{P - C_V} = 3000 \times 450 = 1350000$$

13. In linear programming, constraints can be represented by
    a. equalities
    b. inequalities
    c. ratios
    d. both a and b

14. In Graphical solution the feasible region is _____ .

a. where all the constraints are satisfied simultaneously.

b. any one constraint is satisfied .

c. only the first constraint is satisfied .

d. any one of the above condition.

15. Optimal solution in an LPP is _____.

a. which maximizes or minimizes the objective function.

b. which maximizes the objective function.

c. which minimizes the objective function.

d. which satisfies the non negativity restrictions .

Best wishes